



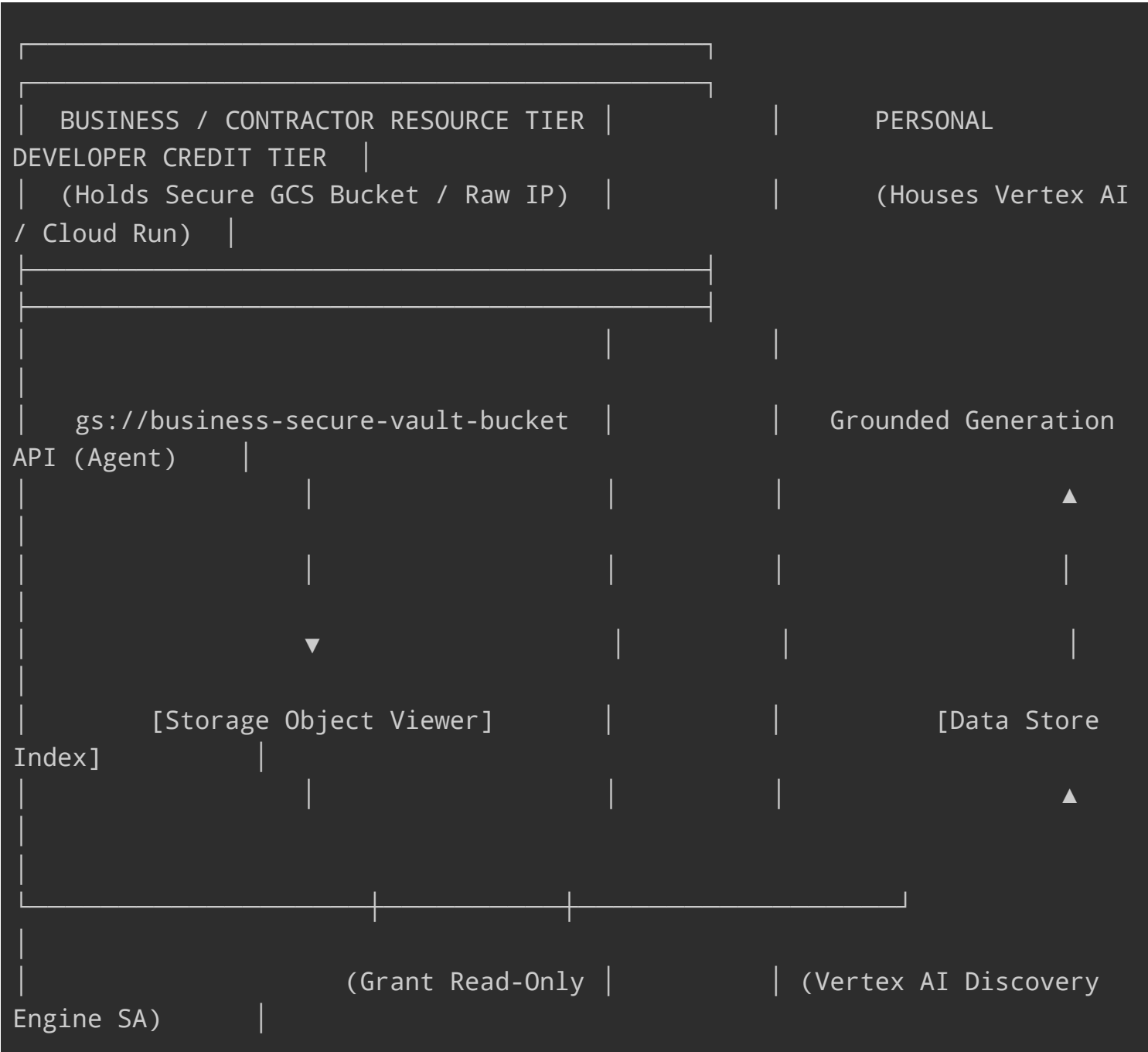
GCP CROSS-ACCOUNT SECURE DATA HANDSHAKE PROTOCOL

Subsystem: Cloud-to-Edge Data Ingestion & Credit-Funded RAG Auditing

Version: 1.0.0 (Production Blueprint)

1. THE ARCHITECTURAL PLAN: DIVISION OF LABORS

To protect your intellectual property behind strict corporate/business workspace walls while absorbing 100% of the heavy query, indexing, and token-processing costs via your personal account's **\$1,000 Vertex AI Agent Builder credit pool**, the system uses an asymmetric, cross-organizational IAM handshake.



Core Design Principles:

1. **No Data Relocation (Zero-Copy):** Raw data files never physically duplicate or migrate into your personal developer environment. They remain inside the business or contractor's encrypted storage vault, maintaining 100% local IP ownership.
2. **Compute-Credit Redirection:** The personal-tier project initiates and manages the indexing, semantic search, and grounding executions. This ensures that **every billing token represents an application application SKU charge** that is absorbed by your personal \$1,000 credit pool rather than hitting business or personal debit cards.
3. **Instant Access Severance:** If a contractor contract ends or employee clearance is revoked, the business administrator simply removes the Service Account member from the bucket's Access Control List (ACL). Access is terminated instantly at the cloud boundary, leaving the consumer project with empty indexes.

2. THE ROUTE MAP & PRINCIPALS

- **Credit-Consumer Project (Personal Account):**
 - *Role:* Executes the RAG pipelines, schedules nightly logs analysis, and hosts the Cloud Run gateway.
 - *Service Account:* `service-PERSONAL_PROJECT_NUMBER@gcp-sa-discoveryengine.iam.gserviceaccount.com` (This is Google's Discovery Engine system identity used to crawl storage paths).
- **IP-Owner Project (Business Account or Contractor):**
 - *Role:* Holds the source files, raw code repository logs, and operational specifications.
 - *Resource:* `gs://business-secure-vault-bucket/` (Standard Google Cloud Storage bucket).
- **The Connection Gate:**
 - *Role:* `roles/storage.objectViewer` (Read-only object access; blocks writing, deleting, or credential modifications).

3. THE HANDOFF BluePrint DOCUMENT (Step-by-Step)

Phase 1: Retrieve Your Personal Project Number

1. Log into the Google Cloud Console using your **personal account** (holding the credits).
2. Ensure you have selected your active developer project.
3. On the Home Dashboard, look at the **Project Info** card and copy the **Project Number** (e.g., 8589934592).

Phase 2: Assemble Your Service Account Identifier

Construct the email identifier for your personal Vertex AI system agent using your copied Project Number:

```
service-PERSONAL_PROJECT_NUMBER@gcp-sa-discoveryengine.iam.gserviceaccount.com
```

Phase 3: Business Bucket Configuration (To be executed by the Business Admin)

The owner of the sensitive files must grant your personal service account read-only access to their bucket:

1. Open the Google Cloud Console using the **Business Workspace account**.
2. Navigate to **Cloud Storage > Buckets** and click on your secure bucket (gs://business-secure-vault-bucket).
3. Select the **Permissions** tab and click **Grant Access** (or **Add Principal**).
4. Paste the Service Account email created in Phase 2 into the **New Principals** field.
5. In the **Select a Role** menu, choose **Cloud Storage > Storage Object Viewer** (roles/storage.objectViewer).
6. Click **Save**.

Phase 4: Create the Data Store in Your Personal Vertex AI Suite

Now, connect the secure bucket directly to your credit-funded Agent Builder:

1. Return to your **Personal Account** GCP Console.
2. Navigate to **Vertex AI Agent Builder > Data Stores**.
3. Click **Create Data Store** and choose **Cloud Storage**.
4. In the Cloud Storage path field, enter the direct address of the business bucket:
gs://business-secure-vault-bucket/
5. Choose **Unstructured documents** (if parsing PDFs, text files, or Markdown notes) and select **Continuous import** to allow real-time changes to stream directly into your local cache indices.
6. Click **Create**. Your Vertex AI agent can now safely parse and ground its responses using your business's proprietary manuals without copying them!

4. THE AUTOMATED HANDOFF BOOTSTRAP SCRIPT

Save this script as `gcp_cross_account_handshake.sh` and make it executable with `chmod +x gcp_cross_account_handshake.sh`. You can run it inside your Termux shell or local CLI environment to automate the setup process using Google Cloud SDK (`gcloud`).

```
#!/usr/bin/env bash
#
=====
====
# AESOP XI: CROSS-ACCOUNT RESOURCE INGESTION & CREDIT HARNESS
# Script ID: AX-GCP-CROSS-ACCOUNT-HANDSHAKE-BOOTSTRAP
#
=====
====
set -euo pipefail

# Style definitions for clean terminal telemetry output
RED='\033[0;31m'
GREEN='\033[0;32m'
CYAN='\033[0;36m'
```

YELLOW='\033[1;33m'

NC='\033[0m' # No Color

clear

echo -e

"\\${CYAN}=====

echo -e "🌐 [GCP CROSS-ACCOUNT HANDSHAKE] Initializing Sovereign
Connection Pipeline..."

echo -e

"=====

Ensure gcloud CLI is installed

if ! command -v gcloud &> /dev/null; then

echo -e "\\${RED}❌ ERROR: Google Cloud SDK (gcloud) is not installed
in this shell.\\${NC}"

echo -e "👉 Install it via: pkg install google-cloud-sdk (or follow
your OS guide)."

exit 1

fi

Step 1: Identify Consumer (Personal Account) Project

echo -e "\n\\${YELLOW}🔑 [STEP 1/3] Fetching Personal Credit-Consumer
Project Details...\\${NC}"

read -p "Enter your PERSONAL GCP Project ID (where the \\\\$1k credits
live): " PERSONAL_PROJECT_ID

Verify active project setup

echo -e "Checking configuration status for project:

\\${CYAN}\\$PERSONAL_PROJECT_ID\\${NC}..."

if ! gcloud config set project "\\$PERSONAL_PROJECT_ID" &> /dev/null; then

echo -e "\\${RED}❌ ERROR: Project ID invalid or not authenticated via
gcloud CLI.\\${NC}"

echo -e "👉 Please run 'gcloud auth login' and try again."

exit 1

fi

Extract the literal Project Number

PROJECT_NUMBER=\\$(gcloud projects list --

filter="projectId=\\$PERSONAL_PROJECT_ID" --format="value(projectNumber)")

if [-z "\\$PROJECT_NUMBER"]; then

echo -e "\\${RED}❌ ERROR: Failed to retrieve project number for

```

\${PERSONAL_PROJECT_ID}.\${NC}"
    exit 1
fi

SERVICE_ACCOUNT="service-\${PROJECT_NUMBER}@gcp-sa-
discoveryengine.iam.gserviceaccount.com"
echo -e "\${GREEN}✅ Success: Personal Project Number verified:
\${PROJECT_NUMBER}\${NC}"
echo -e "Your Personal Vertex AI Service Account is:
\${CYAN}\${SERVICE_ACCOUNT}\${NC}"

# Step 2: Configure Business/Contractor Bucket Access
echo -e "\n\${YELLOW}📁 [STEP 2/3] Configuring Read-Only Access on
Business Resource Bucket...\${NC}"
read -p "Enter the secure GCS Bucket Name (e.g. business-secure-vault-
bucket): " BUCKET_NAME
# Trim gs:// prefix if entered by the user
BUCKET_NAME=\${BUCKET_NAME#gs://}

echo -e "\nTo establish the secure access bridge, choose an execution
strategy:"
echo -e "1) \${CYAN}Execute Now:\${NC} Run the IAM update directly
(Requires being currently logged into gcloud with Business admin
credentials).\"
echo -e "2) \${CYAN}Generate Handoff Block:\${NC} Output a pre-formatted
CLI code block you can copy and email to your business admin /
contractor.\"
read -p "Select option [1 or 2]: " EXEC_MODE

if [ "\$EXEC_MODE" == "1" ]; then
    echo -e "\nApplying policy binding on GCS target bucket:
\${YELLOW}gs://\${BUCKET_NAME}\${NC}..."
    if gcloud storage buckets add-iam-policy-binding "gs://\${BUCKET_NAME}"
\
        --member="serviceAccount:\${SERVICE_ACCOUNT}" \
        --role="roles/storage.objectViewer" &> /dev/null; then
        echo -e "\${GREEN}🎉 SUCCESS: Secure cross-account permission
bridge established successfully!\${NC}"
    else
        echo -e "\${RED}❌ ACCESS DENIED: Failed to modify bucket
permissions.\${NC}"
        echo -e "👉 You may need to use Option 2 and have the Business
Administrator run the command."
    fi
fi

```

```

fi
else
    echo -e "\n\${CYAN}-----
-----"
    echo -e "📋 BUSINESS ADMIN CODES (EMAIL TO CONTRACTOR OR RUN ON
WORKSPACE CONSOLE)"
    echo -e "-----
-----\${NC}"
    echo -e "Copy and execute this exact command block to grant read-only
RAG access:"
    echo -e "\n\${YELLOW}gcloud storage buckets add-iam-policy-binding
gs://\${BUCKET_NAME} \"
    echo -e "  --member=\"serviceAccount:\${SERVICE_ACCOUNT}\" \"
    echo -e "  --role=\"roles/storage.objectViewer\"\${NC}"
    echo -e "\n\${CYAN}Alternative Legacy Utility Command:\${NC}"
    echo -e "\${YELLOW}gsutil iam ch
serviceAccount:\${SERVICE_ACCOUNT}:objectViewer
gs://\${BUCKET_NAME}\${NC}"
    echo -e "\${CYAN}-----
-----\${NC}"
fi

# Step 3: Guidelines for finalization
echo -e "\n\${YELLOW}⚡ [STEP 3/3] Finalizing Data Store
Creation...\${NC}"
echo -e "1. Go to your \${CYAN}Personal Vertex AI Agent Builder\${NC}
console."
echo -e "2. Create a new \${CYAN}Data Store\${NC} using \${YELLOW}Cloud
Storage\${NC}."
echo -e "3. Point the connection directly to:
\${GREEN}gs://\${BUCKET_NAME}/\${NC}"
echo -e "4. Your Vertex AI system is now fully authorized to crawl, parse,
and index your business's files!"
echo -e
"\${CYAN}=====
=====\${NC}"

```